

Addis Ababa University

Introduction to Computer Programming Final Exam for AAiT and EiABC students

Time allotted: 2 hours

Name: _____ ID: _____

Institute: _____ Section: _____

Instructor's name: _____

General Instructions:

- Make sure that the exam contains three parts: multiple choice, write the output of the program, and write code based on the provided description.
- Make sure the exam has 6 pages including the instruction page
- Write your answers on the provided space
- A rough paper will be provided to formulate your answer. Make sure you write your name and id number on it. It will be returned at the end of the exam.
- Any attempt to cheat or refusal to follow instructions will result in severe punishment

For instructor use only

Parts	Part I (15 %)	Part II (15 %)	Part III (20 %)	Total (50%)
Scores				

Part I: Multiple Choice (1.5 points each)

1. Which one of the following data structures is a collection which is ordered and unchangeable, and allows duplicate members?
 - a. Dictionary
 - b. List
 - c. Tuple
 - d. Set
2. Which of the following is true regarding the flow of execution in a while statement?
 - a. Evaluates the condition yielding only True
 - b. If the condition is false, exit the while statement and continue execution
 - c. If the condition is yielding not true, execute the body
 - d. All
3. Which of the following is true?
 - a. Lists are immutable
 - b. Strings are mutable
 - c. All data structures are immutable
 - d. None
4. Which of the following is a permissible variable name?
 - a. _123
 - b. 1_student
 - c. num#1
 - d. except
5. Which of the following is not true about functions in python?
 - a. Functions are reusable pieces of program
 - b. Functions provide better modularity for your programs
 - c. Function definitions can alter flow of execution
 - d. Python allows user defined functions
6. Which of the following function declarations is correct to define a function that accepts a variable number of arguments?
 - a. def func(**args):
 - b. def func(args*):
 - c. def func(*args):
 - d. Not possible in python

7. What is the output of the following statements?

```
my_list = [1,2,3,4,5,6,7,8,9,10]
print(my_list[5::-2])
```

- a. [5,7,9]
- b. [5,4,3,2,1]
- c. [6,4,2]
- d. []

8. What is the output of the following statements?

```
tuple1 = (1, 2, 3, 4)
print(tuple1[1:-1])
```

- a. (1,2,3,4)
- b. (2,3,4)
- c. (2,3)
- d. Error

9. What is the output of the following statements?

```
str1 = "PYTHON programming"
print(str1[2] * str1.index("O"))
```

- a. YYYY
- b. TTTT
- c. THON
- d. OOOO

10. How many times is the print statement executed?

```
for i in range(6):
    for j in range(i):
        print("#")
```

- a. 17
- b. 11
- c. 15
- d. 20

Part II: What is the output of the following snippets (5 points each)

Marking direction:

- 5 marks for writing the exact output
- 3 marks for a reasonable attempt but not getting the exact output
- 1 mark if the answer shows the student understood the question but not a close answer
- 0 mark for no attempt or if the answer shows the student didn't understand the question

1.

```
a = 10
b = 0
while a > b:
    print( "a = ",a, " b = ",b)
    c = a % 3
    d = 5 - b
    for i in range(c , d , 3):
        print( "c = ",c, " d = ",d)
    a -= d
    b+=2
```

Write the output below:

```
a = 10 b = 0
c = 1 d = 5
c = 1 d = 5
a = 5 b = 2
c = 2 d = 3
```

2.

```
rows = 6
string = "PYTHON"
for i in range(rows, 0, -1):
    j = 0
    for k in string:
        if(j==i):
            break
        print(j, end=k)
        j+=1
    print("")
```

Write the output below:

```
0P1Y2T3H4O5N
0P1Y2T3H4O
0P1Y2T3H
0P1Y2T
0P1Y
0P
```

3.

```
my_list = [[1,2,3],[4,5,6],[7,8,9]]
for i in my_list:
    j = 2
    while(j >= 0):
        print(i[j], end = " ")
        j-=1
    print()
```

Write the output below:

```
3 2 1
6 5 4
9 8 7
```

Part III: Write code based on the provided description (10 points each)

Marking direction:

- 10 marks if the provided answer is correct.
- 8 marks if the provided answer has minor syntactic and semantic mistakes but close enough to the correct answer.
- 5 marks if the answer shows the student understood the question and provided a reasonable attempt.
- 2 marks if the student provided a syntactically correct function definition but the statements are not correct.
- 0 marks if the student didn't attempt the question or didn't provide syntactically correct function definition.

1. Write a function that would take a string as an argument cut it down in the middle and inverse it:

Eg. If the function is named my_func then

my_func("abc") returns "bca"

my_func("abcd") returns "cdab"

```
def MidInverse(my_str):  
    length = len(my_str)  
    mid = length//2  
    return my_str[mid:] + my_str[:mid]
```

2. Write a function that accepts a list of integers as an argument and returns the second largest integer in the list.

For eg. If the name of the function is my_function:

my_function([1,3,4,2,5]) returns 4

```
def SecondLargest(my_list):  
    largest = 0  
    second_largest = 0  
    for num in my_list:  
        if num > largest:  
            second_largest = largest  
            largest = num  
        elif largest > num > second_largest:  
            second_largest = num  
    return second_largest
```